Subject:

CIDR :- Classless Inter Domain Region

$$w.x.y.z/\underbrace{s}_{\text{network bits / mask}}$$

eg:-

$$10.09.18.22/\underbrace{26}_{} \rightarrow \text{network bits}$$

$$32-26 = \underbrace{6}_{\text{host bits}}$$

↳ 202.89.69.102 /30

Mask ⇒ 11111111.11111111.11111111.11111100

Q. 202.89.69.102/27

Q. 202.89.69.102/22

Subject:

(1) One of Address of Block is :

201.99.88.79/26

No. of Address in a block

$= 2^{32-x} = 2^{32-26} = 2^6 = \underline{64}$ Address

Mask 11111111.11111111.11111111.11000000

(i) Range of block (iii) Ist Host

(ii) Net Id (iv) Last Host

(v) DBA

N/W

79 → 01001111

Mask → 11000000

01000000 → Net ID

expand by help of mask

$2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 1$

↳ 01000000 → 64

01111111 → 127

Subject:

Net ID :- 201.99.88.64

Ist Host :- 201.99.88.65

Last Host :- 201.99.88.126

DBA → 201.99.88.127

Rules of Classless Address :-

* Address in a block are continuous
* first address of the block should be excatly divisible by number of address of block.

(2) One of the address of Block is :

200.80.90.89 / 27

no. of Addr of Block :-

$$2^{32-27} = 2^5 = 32$$

Mask 11111111.11111111.11111111.111 00000

Range :- $89 \rightarrow \underbrace{010}_{N/W}\underbrace{11001}_{Host}$

$010\ \underline{00}\ 000 \rightarrow 64$

Subject: 010 11111 → 95

64 → 95

One of the Address of block is given

201.99.88.89 / 26 . if this block is divided into 4 equal subblock .

no. of Address in block

$$2^{32-26} = 2^{6} = \underline{64}$$

$$\frac{64}{4} = 16$$

$89 \rightarrow \underline{01}011001$

full range of block

01000000 → 64

01111111 → 127

in 4 so.

16 → 01 00 0000 (64) ↕ 1111 (79)

1, 2, 4, 8 → 15

01 01 0000 (80) ↕ 1111 (95)

01 10 0000 (96) ↕ 1111 (111)

01 11 0000 (112) ↕ 1111 (127)

Subject: # Diffⁿ in supernet & subnet.